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# Enhancing public service delivery through adaptive multichannel management in Northern Tunisia

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**ABSTRACT**

This study investigates how citizens in Northern Tunisia engage with public services through a multichannel management (MCM) framework, integrating both digital and traditional channels. Drawing on survey data from the Beja governorate, it examines how demographic factors, task complexity, and digital access influence citizens' channel choices. Findings reveal that while digital tools are widely available, offline channels remain dominant due to trust, usability concerns, and cultural preferences. Online platforms are primarily used for information-seeking, whereas face-to-face interactions are preferred for complex or high-stakes tasks. The study challenges assumptions of automatic digital migration, highlighting the need for context-sensitive, inclusive service strategies. It contributes to the literature by situating channel behavior within a North African context, offering empirical insights that expand multichannel governance theories beyond the Global North.

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## Introduction

The demand for efficient and accessible public services has intensified globally, particularly in regions facing socioeconomic challenges and digital transformation pressures (Margetts & Dunleavy, 2013; West, 2004). Northern Tunisia presents a compelling case for examining public service delivery mechanisms, characterized by its unique demographic composition and evolving digital landscape. While extensive research has documented citizens' channel selection behaviors and multichannel management (MCM) strategies in developed contexts (Chatfield & Reddick, 2018; Pieterson & Ebbers, 2020), significant gaps remain in understanding these dynamics in North African settings, where cultural norms and infrastructure constraints may yield distinct patterns of channel use.

Recent research has substantially advanced the understanding of channel choice behavior. Numerous studies since 2010 have systematically examined how citizens navigate between digital and traditional service channels based on task complexity, trust factors, and demographic characteristics (Madsen & Kræmmergaard, 2015; Wirtz, 2022). For instance, Pieterson and Ebbers (2020) demonstrated that while digital channels dominate information retrieval, citizens consistently revert to traditional channels for complex transactions, a phenomenon they term 'channel switching behavior'. Similarly, Kernaghan (2015) and Chatefield & Reddick (2018) have established robust frameworks for analyzing channel preferences across different demographic groups (Chatfield & Reddick, 2018; Kernaghan, 2015).

In this study, Multichannel Management (MCM) refers to the strategic integration of diverse service delivery channels including digital platforms, face-to-face services, printed media, and community-based mechanisms to enhance accessibility and citizen satisfaction (Santamaría-Philco & Wimmer, 2018; Wirtz et al., 2023). Channel Choice (CC) is understood as the process by which citizens select a particular service channel based on factors such as task complexity, trust, digital literacy, and accessibility (Pieterson et al., 2017). These definitions guide the analytical framework of this research.

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Contemporary research now recognizes MCM as a mature and strategically adaptive approach, evolving from earlier ambiguities into a widely accepted model for service optimization (Alexopoulos et al., 2019; Wirtz & Birkmeyer, 2015). Rather than debating basic definitions, recent scholarship focuses on channel synergy and integration tailored to local needs (Castilla et al., 2021). This study builds upon this consensus while emphasizing the need for localized, context-sensitive applications in regions such as Northern Tunisia.

In the Tunisian context, where digital infrastructure coexists with strong traditions of face-to-face governance, examining channel preferences offers valuable insights. While global patterns of channel use are well-documented (Madsen & Kræmmergaard, 2016), Tunisia's particular combination of high mobile penetration (92% smartphone ownership) and persistent offline service preferences presents an important opportunity to test and refine existing MCM theories. This study builds on established channel choice literature while addressing the critical gap in North African empirical research, aiming to contribute both to academic understanding and practical improvements in public service delivery.

Northern Tunisia presents a distinctive case due to its sociocultural and infrastructural landscape, where service delivery is shaped by uneven internet connectivity, strong traditions of relational governance, and varying levels of digital literacy. Unlike Northern Europe or North America, where multichannel management has matured within high-trust, digitally integrated societies (Abdelsalam et al., 2013; Verhoef et al., 2005), Tunisia's service ecology reflects a hybrid system in transition. The reliance on offline channels among even technologically capable individuals points to a complex interplay of access, trust, and usability, reinforcing calls for context-sensitive multichannel strategies (Criado & Villodre, 2018; Klievink et al., 2017). Our findings extend existing models by situating channel behavior within a North African context that resists universalist assumptions.

Despite extensive literature on multichannel behavior in digitally mature societies, there remains a critical lack of empirical research in North African contexts, where digital adoption is uneven and trust in institutions varies. This study addresses that gap by examining the interplay between channel choice, user demographics, and task complexity in Northern Tunisia.

The primary objective of this study is to evaluate how citizens in a semi-digitized public service environment choose between online and offline channels and to identify the factors influencing these decisions.

By analyzing channel preferences across demographic groups and types of public service needs, this article contributes to the literature by offering a localized perspective that challenges universal models of digital service adoption and emphasizes the importance of cultural and infrastructural factors in multichannel management.

The remainder of this paper proceeds as follows: First, we review the literature on multichannel management and channel choice behavior, with particular attention to recent theoretical advances. Next, we present our methodology for data collection in Northern Tunisia, followed by an analysis of channel preferences across different demographic groups and service types. Finally, we discuss implications for both theory and practice, highlighting how our findings compare with and extend existing research in the field.

## Literature review

The study of multichannel management (MCM) and channel choice (CC) in public service delivery has undergone significant transformation over the past three decades. What began as an optimistic belief in the dominance of digital channels has evolved into a nuanced understanding of how multiple service channels interact, influenced by technological, organizational, and user-related factors. This literature review traces the development of MCM and CC research, highlighting key theoretical contributions, empirical findings, and emerging trends while identifying gaps for future inquiry.

### *Early optimism and the emergence of multi-channeling (1990s–Early 2000s)*

The advent of the Internet in the 1990s sparked widespread anticipation that digital channels would revolutionize service delivery across both public and private sectors. Scholars and practitioners

predicted that online platforms would replace traditional face-to-face and physical channels due to their advantages in accessibility, speed, and cost-efficiency (Castilla et al., 2021). In the private sector, businesses rapidly adopted e-commerce models, expecting digital channels to become the primary means of distribution (Bélanger & Carter, 2008; Dijk & Hacker, 2018). Similarly, in the public sector, early e-government initiatives promoted online services as a way to enhance efficiency and citizen engagement (Moon, 2013).

However, by the early 2000s, it became clear that digital transformation alone could not address the diverse needs of all users. Research revealed that while online channels were effective for simple transactions, many citizens and customers still preferred traditional methods for complex or sensitive interactions (Riggins & Mitra, 2007). This realization led to the emergence of multi-channeling, where organizations provided services through multiple parallel channels online, phone, and in-person rather than relying on a single dominant medium (Vries, 2020).

During this period, scholars began examining the factors influencing channel preferences. Lee (2002) found that demographic characteristics such as age, education, and gender significantly shaped individuals' choices, reinforcing the need for diverse service delivery options (Lee, 2002). Meanwhile, Verhoef et al. (2005) developed foundational theories comparing online and offline channels, demonstrating that each had distinct strengths: digital platforms excelled in convenience, while physical interactions provided trust and personalization (Verhoef et al., 2005). These insights laid the groundwork for more sophisticated MCM frameworks in subsequent years.

### *Theoretical advancements and sector-specific insight (mid-2000s–2010s)*

As digital adoption grew, research on MCM and CC became more refined, moving beyond descriptive studies to develop structured models of channel selection and integration. A pivotal contribution came from Pieterson (2010), who emphasized that multi-channel strategies should not merely offer parallel options but strategically align channels with specific service requirements (W. Pieterson, 2010). This perspective was further developed by Wirtz and Langer (2017), who introduced a comprehensive MCM framework identifying four key dimensions: channel characteristics, user segmentation, organizational integration, and performance measurement. Their work highlighted the importance of coordination, rather than mere coexistence of channels, a concept later expanded upon by Santamaria-Philco et al. (2019) under the label of channel synergy (Santamaria-Philco et al., 2019).

In the public sector, studies by Reddick (2005) and Thomas and Streib (2003) revealed persistent gaps between digital service availability and actual usage. Citizens frequently turned to government websites for information but preferred phone or in-person interactions for complex tasks, particularly when trust or clarity was a concern (Thomas & Streib, 2003). These findings challenged earlier assumptions that digital channels would naturally supplant traditional ones and underscored the need for flexible, user-centric approaches. Millard (2007) formalized this idea with the concept of **'**flexi-channeling**'**, distinguishing between **'**single-channelers**'** (those reliant on one method, typically in-person) and **'**flexi-channelers**'** (those who dynamically switched between channels based on need) (Millard, 2007).

Meanwhile, private-sector research provided valuable comparative insights. While businesses increasingly prioritized cost-effective digital platforms, sectors like banking retained physical branches for high-stakes transactions, illustrating that channel optimization required balancing efficiency with user preferences (Vives, 2019). These lessons informed public-sector strategies, particularly in e-government, where scholars like Pina et al. (2009) and Yildiz (2007) documented the enduring role of face-to-face services despite digital advancements (Pina et al., 2009; Yildiz, 2007).

### *Maturation of MCM research: integration, context, and equity (2010s–present)*

The past decade has seen MCM scholarship mature, with a stronger emphasis on empirical validation, contextual factors, and equity considerations. Pieterson et al. refined their earlier models by incorporating new evidence on channel-switching behaviors, identifying three core determinants of CC: task characteristics (e.g. complexity, urgency), user characteristics (e.g. digital literacy, trust), and channel characteristics (e.g. accessibility, privacy) (Pieterson & Ebbers, 2020). This tripartite model has been

validated across diverse settings, including developing nations (Chatfield & Reddick, 2018) and specialized service domains (Lindgren et al., 2019).

A significant development has been the recognition of national and cultural influences on channel preferences. Comparative study of Egypt and Canada demonstrated that service complexity affected channel choice differently depending on institutional trust and digital infrastructure (Abdelsalam et al., 2013). Similarly, Klievink et al. (2017) highlighted how intermediary organizations (e.g. libraries and community centers) shaped channel use patterns, particularly among digitally excluded groups (Panagiotopoulos et al., 2019). These findings challenged the notion of a universal channel hierarchy and underscored the need for context-sensitive MCM strategies (Pieterson et al., 2017).

Parallel advancements in digital divide research further enriched MCM scholarship. Early studies focused on access gaps (van Dijk et al., 2007), but contemporary work emphasizes usage and outcome disparities (van Deursen & Helsper, 2015). For instance, Chen and Dimitrova (2006) showed that even when channels are technically available, marginalized groups may underutilize them due to skill gaps or mistrust (Chen & Dimitrova, 2006). Ebbers et al. (2020) addressed this through digital mediation strategies, such as training programs and hybrid service designs, which bridge gaps between online and offline users (Pieterson & Ebbers, 2020).

Recent systematic reviews have consolidated these insights. Madsen and Kræmmergaard's (2016) analysis of 127 CC studies established robust typologies of choice determinants, while Santamaria-Philco et al. (2019) synthesized evidence on channel synergy, arguing that optimal MCM requires active coordination rather than passive coexistence. Additionally, behavioral insights have gained traction, with Pieterson and Ebbers (2020) incorporating nudge theory and habit formation into CC models to explain why users often default to suboptimal channels.

This literature shows that the evolution of MCM and CC research reflects a broader shift from techno-optimism to evidence-based, user-centric approaches. Early expectations of digital dominance gave way to recognition of the complex interplay between technology, organizational structures, and human behavior. Today, the field emphasizes strategic integration, equity, and adaptability, offering valuable insights for policymakers seeking to optimize public service delivery. However, as digital landscapes continue evolving, future studies must explore AI-driven channel management, longitudinal behavior patterns, and inclusive design principles to ensure MCM systems serve all citizens effectively. Here, after a summary of the key findings of the literature review, a chronological timeline table is presented (Table 1).

**Table 1.** Literature review summary.

| Time Period | Key Themes | Major Contributions | Implications |
|---|---|---|---|
| *1990s–2000s (Digital Optimism)* | • Digital optimism.<br>• Emergence of multi-channeling.<br>• Recognize limitations. | • Castilla et al. (2021): Internet's potential.<br>• Riggins and Mitra (2007): Digital limitations.<br>• Thomas and Streib (2003): Face-to-face preference. | • Need for complementary channels.<br>• Public sector adopts private-sector concepts. |
| *2000s–2010s (Frameworks & Insights)* | • Theoretical frameworks<br>• Behavioral insights<br>• Demographic disparities<br>• Municipal digital platforms | • Verhoef et al. (2005): Channel comparisons.<br>• Pieterson et al. (2017): Tripartite model.<br>• Millard (2007): 'Flexi-channelers' concept.<br>• Wirtz and Kurtz (2016): Social media in governance. | • Inclusive MCM strategies required.<br>• Digital divide extends beyond access. |
| *2010s–2020s (Integration & Context)* | • Strategic synergy.<br>• Contextualization.<br>• Digital divide reframed.<br>• Usage/outcome disparities. | • Wirtz and Langer (2015): Four-pillar framework.<br>• Reddick et al. (2012): Cross-national studies.<br>• Helsper (2012): Second-level digital exclusion.<br>• Ebbers et al. (2020): Digital mediation. | • Context-sensitive frameworks needed.<br>• Equity requires assisted services. |
| *2020s–Present (Contemporary Trends)* | • Channel synergy.<br>• Behavioral insights.<br>• Smart cities.<br>• Co-creation. | • Santamaria-Philco et al. (2019): Synergy frameworks.<br>• Pieterson and Ebbers (2020): Behavioral economics. | • Dynamic, adaptive systems.<br>• Participatory design. |
| *Future Directions* | • Policy impact.<br>• Digital inclusion.<br>• Behavioral drivers. | • Madsen and Kræmmergaard (2016): Migration gaps.<br>• Van Deursen et al. (2015): Outcome disparities. | • Interdisciplinary research.<br>• Localized strategies. |

## Hypothesis

Research indicates that citizens use a variety of channels to communicate with government authorities, a phenomenon driven by the need for flexibility and accessibility in accessing services. According to Pieterson and Ebbers (2008), the choice of service delivery channels reflects differences in task complexity, personal preferences, and accessibility (Pieterson & Ebbers, 2008). Citizens often interchange between digital and offline channels based on the nature of the service required, as observed by Reddick (2005), who found that complex tasks are more frequently resolved using personal interaction, while information retrieval is commonly done online (Reddick, 2005).

Additionally, Millard (2007) emphasizes the concept of 'flexi-channeling', where citizens demonstrate adaptability in their channel usage, selecting based on convenience, familiarity, and trust. Lee (2002) also suggests that age, education, and technological comfort influence channel preference (Lee, 2002; Millard, 2007). These studies prove that citizens actively use multiple channels to engage with government services.

Therefore, to determine whether people use numerous channels (**'**multi (or flexi) channelers**'**), the first hypothesis is formulated:

**H1:** Citizens use many channels to communicate with government authorities.

A deeper examination of the factors influencing citizens' preference for offline channels over online ones reveals a multifaceted issue beyond mere access to digital devices. Fundamental to this is the **'**digital divide**'**, a concept introduced by Cai (2008), which highlights not only the disparity in physical access to technology but also the proficiency in utilizing these tools effectively (Cai, 2008). Despite the proliferation of connected devices, many citizens lack the necessary digital literacy, inhibiting their ability to leverage online channels for interacting with government services. Moreover, Chen & Dimitrova (2006) suggest that demographic characteristics such as age, education, and income levels significantly impact individuals' internet usage patterns. (Chen & Dimitrova, 2006).

However, it is essential to note the complex interplay of these factors; while some studies cite gender as a determinant of online engagement, findings are inconsistent, reflecting the need for nuanced analysis. This complexity is mirrored in Helsper's (2012) research on digital inclusion, demonstrating how socioeconomic and behavioral barriers often shape preferences for offline communication methods (Helsper, 2012). For many, traditional face-to-face interactions inherently offer a sense of trust and reliability that digital interfaces struggle to provide. Li et al. (2019) further argue that trust and perceived ease of use are crucial in channel selection, with offline channels often seen as safer and more straightforward. This perception is particularly pervasive among populations with limited exposure to digital interfaces, reaffirming the preferential use of more familiar offline methods (Li et al., 2019).

These insights underscore the necessity for a comprehensive approach to addressing the digital divide. They suggest that government initiatives should focus on increasing access to technology, enhancing digital literacy, and fostering trust in online systems. These factors can significantly aid in strategizing effective multichannel management that resonates with citizens' diverse needs and preferences.

**H3:** Men use online channels more frequently than women.

**H4:** Younger citizens use online channels more regularly than older people.

**H5:** More educated people are likely to use online channels than less educated people.

Research suggests that the specific characteristics of a task can significantly impact the channels citizens choose when interacting with government services. For instance, Noble et al. (2005) and Verhoef et al. (2005) illustrate that simple transactions, like information retrieval or simple inquiries, are often more efficiently handled online due to speed and convenience. Conversely, more complex tasks, such as obtaining permissions or completing legal transactions, might require the depth of understanding and personal interaction that offline channels offer (Noble et al., 2005; Verhoef et al., 2005).

Furthermore, Goldsmith and Crawford (2014) argue that the perceived complexity of a task can lead citizens to seek face-to-face assistance, reflecting a preference for responsiveness and nuanced understanding that digital platforms often lack (Goldsmith & Crawford, 2014).

Further, Venkatesh et al. (2003) suggest that the perceived ease of use and usefulness of a particular channel can steer user behavior, indicating that government services should enhance digital interfaces to meet citizen preferences better and optimize service delivery (Venkatesh et al., 2003).

Aligning these insights with the context underscores the need for versatile service delivery strategies that cater to varying task complexities and user preferences, fostering a more inclusive approach to government- citizen interactions.

**H6:** Citizens choose online channels to seek information.

**H7:** Citizens choose offline channels to carry out procedures, such as making a transaction.

Proximity and direct interaction remain crucial factors influencing citizens' preference for offline channels. The research underscores that face-to-face communication is often perceived as more reliable and empathetic than digital interactions, providing a personal touch that fosters trust (Goldsmith & Crawford, 2014). Furthermore, Villodre & Criado (2020) reveal that complexities in navigating digital platforms can discourage online channel usage, especially among users with limited digital literacy (Villodre & Criado, 2020). Therefore, citizens might gravitate towards offline channels for services requiring nuanced understanding or reassurance. Additionally, trust and familiarity aspects influence this preference, as citizens often rely on personal interactions for assurance in service delivery processes (Li et al., 2019).

These insights highlight governments' importance in maintaining robust offline service options while improving digital access and literacy initiatives.

**H8:** Citizens prefer offline channels over online channels.

## Methodology

This study employs a face-to-face survey methodology to ensure inclusivity across digital literacy levels and device ownership. This approach is particularly relevant in regions like Northern Tunisia, where digital inequalities persist despite high mobile penetration (Chen & Dimitrova, 2006; Helsper, 2012). Face-to-face methods are recognized for improving response rates and data quality in populations with variable ICT access (Tourangeau et al., 2013). The selection of the Beja governorate, currently piloting multichannel municipal services, offers a strategic case for studying citizen interaction across channel types. The sample includes both digitally connected and disconnected individuals, allowing the study to capture channel preferences beyond digitally engaged subgroups—a critical gap identified in multichannel research (Reddick, Abdelsalam and Elkadi, 2012; van Dijk, 2007). The combination of frequency distribution, OLS regression, and non-parametric tests aligns with best practices in public service behavior research where data distributions vary and variables include both ordinal and categorical measures (Field, 2013). This mixed-method quantitative design thus ensures analytical robustness while grounding results in the realities of a hybrid governance system.

The research was conducted in the Beja governorate in northeastern Tunisia (Figure 1). This governorate encompasses eight municipalities, four of which currently offer both online and offline municipal services as part of a national trial program designed to assess electronic service delivery. This initiative is scheduled to operate from 2022 to 2026, with plans for expansion to more communities. The face-to-face survey gathered 259 genuine responses acceptable for analysis. A total of 123 replies were collected from the four municipalities that provide both offline and online services.

The current research was intentionally designed to involve people who do not use connected devices, as they still comprise a sizable proportion of the population. As a result, the questionnaire was administered through face-to-face interviews. Citizens who agreed to participate were asked about their use of smartphones, laptops, internet data plans, and home internet access.

The sample consisted of 48% female and 52% male participants, with an average age of 39. In 46% of cases, the respondents had a university education, whereas in 40% of cases, they had a high school education. The data shows a significant gender gap, with males earning 69% more than females, exceeding the minimum wage of 150 USD. Ninety-two percent of respondents possessed a smartphone with a data plan, and there was no notable difference in this regard across genders. Approximately 24% of the participants demonstrated infrequent use of ICT, while the remaining 76% had regular to frequent use.




**Figure 1.** Tunisia and Beja governorate maps.
Source: https://www.worldatlas.com/maps/tunisia, 2024.

Additionally, 68% of the respondents reported possessing advanced ICT abilities. Half of the sample population uses computers or laptops to connect to the internet. Moreover, 64% of the respondents expressed satisfaction with three or more offline municipal services.

Most variables used for analysis were ordinal (such as age and education) or nominal (such as gender and channel used). Channel satisfaction is measured using a 5-point Likert scale, ranging from 1 to 5. A Frequency distribution test was used to evaluate Hypothesis 1 and 2. Ordinary least square (OLS) regression was used to verify hypotheses 3, 4, and 5, which were questioned or measured on nominal and/or ordinal scales. The Wilcoxon test for paired observations was used to evaluate hypotheses 6 and 7, while the t-test was employed to examine hypothesis 8. Every hypothesis test has a 0.05 alpha.

In the study, respondents were asked to indicate whether they utilized each channel in their interactions with their municipality. The data shows that people interact with government agencies differently. A new variable was introduced to confirm H1 statistically. At first, the questions were nominal (yes/no). The number of channels used was determined by adding the points assigned to each variable to get a score. This modified variable could take on values between 0 (no channels utilized) and 5 (all channels used). A single channel was indicated by one, and several channels were suggested by more than one.

The study used a frequency distribution test to determine whether the observed values were more significant. During the six-year period from 2017 to 2022, 73.28 percent of respondents reported using multiple channels, while 26.73 percent reported using zero to one channel (Figure 2). Approximately 89.65% of observed values were more significant than one ($p < 0.000$), supporting the hypothesis that people use multiple channels. People typically employ three channels (3.02) when communicating with government institutions.

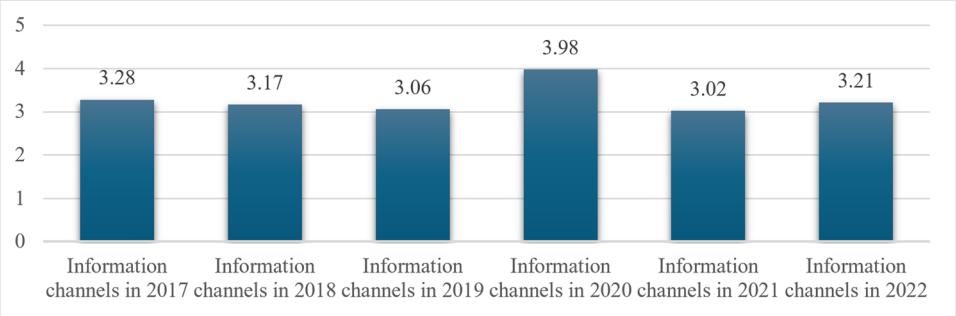



**Figure 2.** Information channel usage means from 2017 to 2022.

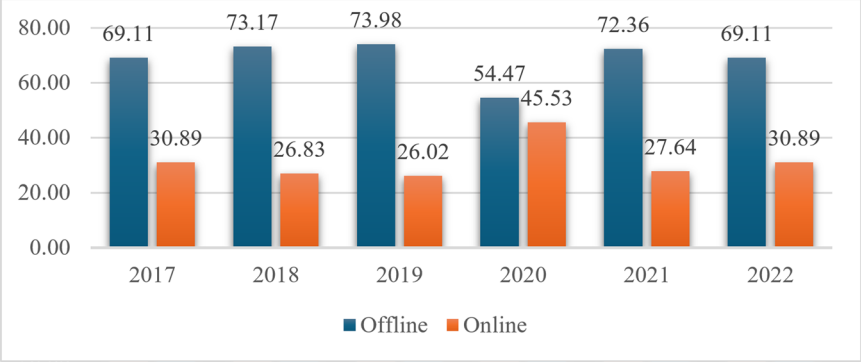



**Figure 3.** Most often used methods for communicating with government agencies.

To check H2 to H5, utilized channels were divided into two groups. Front desk, in-person meetings, printed communication, and car speaker patrols were recorded as offline channels, whereas websites, broadcast media, and social media were recorded as online channels.

For the second hypothesis, respondents were asked about the channel categories (online or offline) they most often used while dealing with government entities (Figure 3).

Offline channels are more commonly cited; 91.86% of respondents own a smartphone with a data plan, and 69.91% are reported as frequent internet users. Approximately 22.76% had only a laptop or computer connected to the internet at home. Moreover, only 19.51% had a smartphone and a connected laptop or desktop. As seen in Figure 3, offline channels were used more frequently than online ones. The distinction between those who own a connected device and those who do not is noticeable. However, face-to-face communication was used substantially more by people who do not own a computer than other channels.

For hypotheses 3, 4, and 5, the study investigated the effects of gender, age, education, and ICT level on channel use. Respondents' ages were categorized into three groups: those under 29, those aged 30–59, and those over 60. Educational levels were categorized into low, medium, and high education. The ICT level is classified into skilled or non-skilled categories.

Table 2 shows that gender and age do not significantly influence channel usage ($p=0.494$, $p=0.183$). At the same time, education and ICT levels do ($p=0.0236$, $p=0.0517$). This means that men and women use offline and online channels equally. Younger respondents tend to use online channels; similarly, better-educated individuals also prefer them.

In examining H3, H4, and H5, the study compared those who owned a connected device to those who did not. Results show no interaction effect between connected device ownership, age, or gender on channel use (Table 3). This indicates that, even after accounting for connected device ownership, there is no difference in channel usage between men and women; however, age and education disparities persist. The three demographic characteristics could potentially interact. For example, young men's preferences will likely differ from older women's. The research utilized a binomial logistic regression analysis to investigate the effects of the interaction. The independent variable was channels (offline vs online),

**Table 2.** Verification of H3 to H5.

| Model 11: Logistic, using observations 1–123 | | | | |
|---|---|---|---|---|
| Dependent variable: Information channel type<br>yhat =~ E(100/(1 + exp(−X*b))) | | | | |
| | *Coefficient* | *Std. Error* | *t*-Ratio | *p*-Value |
| const | −4.05261 | 0.0800634 | −50.62 | <0.0001*** |
| Age | 0.0546161 | 0.0408429 | 1.337 | 0.1837 |
| Gender | −0.0279507 | 0.0408275 | −0.6846 | 0.4949 |
| Education | 0.0738166 | 0.0321866 | 2.293 | 0.0236** |
| ICT_level | −0.0862458 | 0.0438733 | −1.966 | 0.0517* |
| Statistics based on the transformed data: | | | | |
| Sum squared resid | 5.936740 | S.E. of regression | | 0.224302 |
| R-squared | 0.088708 | Adjusted R-squared | | 0.057817 |
| F(4, 118) | 2.871628 | *p*-Value (*F*) | | 0.025974 |
| Log-likelihood | 11.87854 | Akaike criterion | | −13.75709 |
| Schwarz criterion | 0.303834 | Hannan-Quinn | | −8.045577 |
| Statistics based on the original data: | | | | |
| Mean dependent var | 1.878049 | S.D. dependent var | | 0.328568 |
| Sum squared resid | 99.73438 | S.E. of regression | | 0.919351 |
| Log-likelihood | −60.64870 | Akaike criterion | | 131.2974 |

**Table 3.** Verification of H3 to H5.

| Model 13: Logistic, using observations 1-123 | | | | |
|---|---|---|---|---|
| Dependent variable: Information channel type<br>yhat =~ E(100/(1 + exp(−X*b))) | | | | |
| | *Coefficient* | *Std. Error* | *t*-Ratio | *p*-Value |
| const | −4.27606 | 0.150551 | −28.40 | <0.0001*** |
| Age | 0.00324491 | 0.00192573 | 1.685 | 0.0947* |
| Gender | −0.0278825 | 0.0405904 | −0.6869 | 0.4935 |
| Education | 0.0368422 | 0.0169258 | 2.177 | 0.0315** |
| ICT_level | −0.0972450 | 0.0456594 | −2.130 | 0.0353** |
| Mobile_phone_connection | 0.132247 | 0.0889990 | 1.486 | 0.1400 |
| Home_IT_hardwear | 0.0485255 | 0.0412653 | 1.176 | 0.2420 |
| Statistics based on the transformed data: | | | | |
| Sum squared resid | 5.785775 | S.E. of regression | | 0.223332 |
| R-squared | 0.111881 | Adjusted *R*-squared | | 0.065944 |
| F(6, 116) | 2.435533 | *p*-value(*F*) | | 0.029669 |
| Log-likelihood | 13.46266 | Akaike criterion | | −12.92532 |
| Schwarz criterion | 6.759970 | Hannan-Quinn | | −4.929204 |
| Statistics based on the original data: | | | | |
| Mean dependent var | 1.878049 | S.D. dependent var | | 0.328568 |
| Sum squared resid | 99.38665 | S.E. of regression | | 0.925625 |
| Log-likelihood | −59.06459 | Akaike criterion | | 132.1292 |

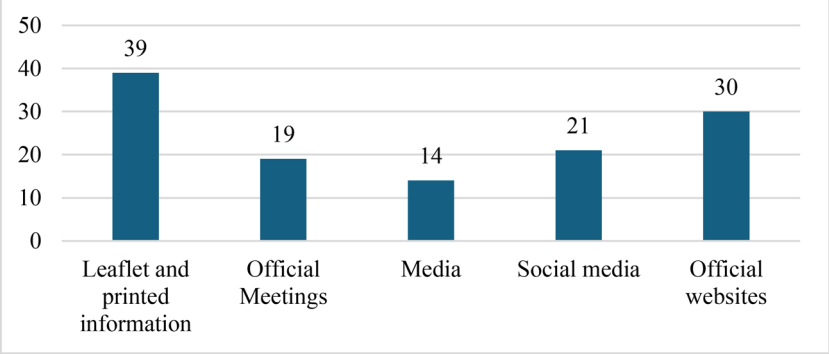



**Figure 4.** Preferred information channels.

whereas the dependent variables were age, education, and gender. This test shows that the collected data (education and ICT level) accurately predicts the preference for an online over an offline channel.

To validate the sixth hypothesis, respondents were inquired about the channels they would utilize to seek information. Figure 4 indicates that leaflets and printed materials are favored, followed by government websites and social media. To examine the seventh hypothesis, respondents were inquired about

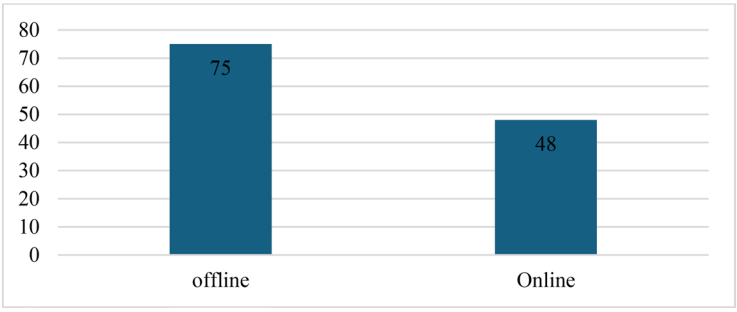



**Figure 5.** Transactions carried out.

**Table 4.** Test for difference between Information channel type and participating online platform.

| |
|---|
| Wilcoxon Rank-Sum Test |
| Null hypothesis: The two medians are equal. |
| n1 = 123, n2 = 123 |
| w (sum of ranks, sample 1) = 22395 |
| $z$ = (22395 - 15190.5) / 558.037 = 12.9104 |
| $P(Z > 12.9104) = 0$ |

the channel they utilized for carrying out an action. Figure 5 demonstrates that offline channels were predominantly used in comparison to internet channels.

To determine whether a preferred channel exists, the study assessed the two channels for all conditions using a Wilcoxon test for paired observations. The online channel was the preferred information search channel ($Z = 12.910$, $p < 0.000$) (Table 4).

Hypothesis height was tested by comparing offline and online channels' intention to use ratings. Offline channels received an average rating of 2.79 out of 5 compared to 2.25 for online channels. The paired-sample $t$-test (Figure 6) indicates a statistically significant disparity between the two channel types ($t(244) = -3.747$, $p < 0.0002$). Likewise, differences exist between people who own a connected device and those who do not. The latter prioritized offline channels, with an average rating of 3, compared to online intention to use, which had an average rating of 2.33. Similarly, people with a connected device intended to use online channels when looking for information on municipalities and services.

## Analysis

In examining how citizens interact with public authorities, data reveals that a significant number of individuals engage through multiple channels, identifying them as multi-channel or flexi-channel users (H1). This observation aligns with prior academic research, demonstrating that citizens utilize a variety of channels to fulfill different needs. Notably, there seems to be no strong inclination towards preferring online services over traditional methods. Consistent with Hypothesis 2 (H2), citizens frequently resort to offline channels, such as counter services and printed documents, alongside online platforms.

The study examines various demographic and socioeconomic factors to investigate the differences observed in channel usage. Specifically, the findings under Hypothesis 3 (H3) indicate that gender does not significantly influence channel preference; both men and women utilize offline and online channels in similar proportions. However, considerable differences emerge concerning age (H4) and educational level (H5). The data consistently shows that age plays an essential role; older individuals are more inclined towards offline channels, whereas younger respondents, including those between 30 and 59, still predominantly rely on offline methods.

Education level and Information and Communication Technology (ICT) competence also emerge as key determinants. Individuals with higher education levels or advanced ICT skills are more likely to prefer online channels compared to those with lower levels of education or less technological expertise. Those with a less educational background or lower ICT skills tend to gravitate significantly more towards offline channels than individuals with moderate to high levels of education and ICT skills. Despite this, it is worth noting that the majority of respondents, regardless of age or education, prefer offline channels, with a preference rate of

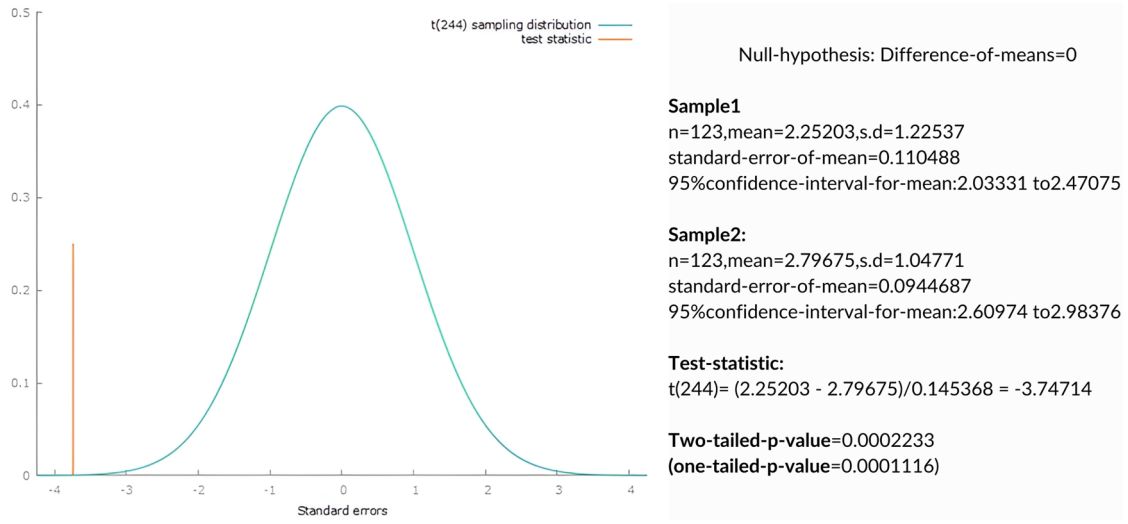



**Figure 6.** Paired-sample *t*-test for comparing offline and online channels' intention to use.

71.54% among those with medium education and 8.94% among those with higher education. This suggests that offline channels are more popular than online alternatives across various demographics.

Recognizing that variations in channel usage are not entirely based on demographic factors, it is essential to investigate why citizens choose different channels or why offline options are preferred despite online availability. The research addressed these concerns by examining two key hypotheses focused on task complexity and problem resolution. The data suggests that citizens choose distinct channels for varying tasks. Straightforward tasks, such as information seeking, are often performed online, while more complex administrative procedures, like form completion, are typically conducted offline.

These findings confirm Hypotheses 6 and 7, highlighting a behavioral pattern whereby citizens primarily use online channels for informational purposes and offline channels for actions. This aligns with earlier studies, reinforcing the understanding that citizens employ multiple channels depending on the task at hand. This suggests a strong need–channel relationship that is particularly context-sensitive in Northern Tunisia. Our results reveal a nuanced relationship between citizen needs and channel choice in Northern Tunisia. Simple tasks, such as information retrieval, are predominantly conducted online, while complex administrative actions prompt a shift to offline interactions. This aligns with Venkatesh et al. (2003), who link perceived task complexity to technology use (Venkatesh et al., 2003).

However, in Tunisia's context, even tech-literate users revert to traditional channels for tasks involving ambiguity or trust, underscoring a culturally embedded preference for interpersonal interaction (Yao & Xu, 2022). Offline channels offer reassurance and clarity that digital interfaces currently lack, particularly in municipalities where digital services are newly introduced and inconsistently supported. Moreover, the strong preference for face-to-face interaction, even among younger and educated populations, suggests that the relational nature of governance in Tunisia deeply influences how needs are mapped onto channels. Thus, task complexity alone cannot predict channel choice without accounting for trust, familiarity, and service maturity within the local governance structure.

Ultimately, while citizens utilize a range of channels, there is a clear preference for offline interactions, which are perceived as more favorable (H8). The study also identifies variations between individuals with access to a connected device and those without. Those without a connected device tend to use fewer channels and hold a more favorable view of offline channels compared to their online-engaged counterparts. This analysis significantly enhances our understanding of channel preferences in citizen- public authority interactions. It underscores the need for strategies that bridge the digital divide while catering to diverse user preferences.

## Discussion

The results from this study partially support the initial hypothesis, indicating that citizens engage with online channels for both fundamental and complex tasks. However, this observation contrasts with other

findings in the literature, such as those by Gamboa Mora et al. (2021), which emphasize that citizens can exhibit consistent patterns across various channels (Gamboa Mora et al., 2021). Consequently, the seventh hypothesis requires further refinement. To achieve this, it is essential to implement a more robust operationalization of problem-solving frameworks, with a particular focus on key determinants such as task difficulty, complexity, and ambiguity. Further research is needed to explore the dynamics that influence citizens' preferences for online channels over traditional in-person interactions, such as town hall meetings.

Goldsmith and Crawford (2014) highlight that the perceived uncertainty experienced by citizens is a crucial factor in shaping their channel choices (Goldsmith & Crawford, 2014). It is essential to evaluate this uncertainty systematically rather than relying solely on citizen preferences. This approach advocates for a more comprehensive assessment of both subjective and objective factors influencing citizen choices, aligning the analysis with theoretical models that account for variations in digital engagement and usability.

To better understand the nuanced distinctions between preference and actual behavior in the context of multi-channel public service delivery, it is essential to delve deeper into the complexity and ambiguity surrounding this area. As preferences do not always translate into concrete actions, future research must emphasize precision in defining and measuring these aspects, as suggested by Venkatesh et al. (2003). Additionally, by leveraging insights from technology adoption theories, such as the Technology Acceptance Model, researchers can gain a clearer perspective on how complexities influence the decision-making process between offline and online channels (Venkatesh et al., 2003). This approach enables a more comprehensive understanding of the factors influencing citizens' channel choices, thereby contributing to the optimization of multi-channel strategies in public administration.

The results, however, validated the assumptions that citizens support numerous channels and, most importantly, that although many services are available online, offline channels are still widely used in many categories. The research finds that online channels are gaining popularity on their own. This result contributes to the scientific understanding of multi-channel management in the public sector, which has emerged in recent years. Still, it also means governments will eventually have to review their current multi-channel management plans. Evidence suggests that many governments believe citizens will automatically prefer online channels over offline ones merely because they offer online services.

To enrich the analysis of multichannel management practices in Northwest Tunisia, it is imperative that government entities consider a holistic approach towards optimizing and integrating communication channels. Integration should not merely emphasize the inclusion of multiple channels but should focus on ensuring seamless interoperability and strategic cross-channel referrals. Such an approach aligns with the concept of channel synergy, which underscores the potential for enhanced efficiency and cost-effectiveness when disparate channels collaborate toward common objectives (Santamaria-Philco et al., 2019). Implementing channel synergy can aid in reducing the inherent complexities and redundancies encountered by citizens when interacting with government services.

Moreover, offline channels' ability to provide citizens with personalized interactions and nonverbal communication can be leveraged against the backdrop of rising digital channels (Yao & Xu, 2022). The latter, with its vast information storage capabilities, serves as an invaluable resource in demystifying complex bureaucratic processes. As corroborated by Heinhuis and De Vries (2009), the efficacy of online platforms can be amplified when complemented by offline channels, offering a human touch in instances marked by ambiguity or uncertainty (Heinhuis & De Vries, 2009).

While our findings align with established literature that citizens prefer online channels for simple, low-stakes tasks and revert to traditional methods for complex interactions (W. Pieterson & Ebbers, 2008; Reddick, 2005), our study adds contextual depth by examining these behaviors in Northern Tunisia a region where digital access does not straightforwardly translate to digital preference. Despite high smartphone penetration, offline channels remain dominant, even among university-educated and digitally skilled respondents, challenging assumptions drawn from studies in digitally mature contexts (Chen & Dimitrova, 2006; B. Wirtz & Kurtz, 2016). This suggests that infrastructural readiness alone cannot drive digital adoption. Instead, the continued reliance on face-to-face communication reflects a need for interpersonal trust and institutional reassurance, especially in transactions perceived as high-stakes elements that are often underemphasized in channel choice models developed in the Global North.

To support the effectiveness of multichannel strategies, governments should also adopt an integrated service delivery framework. This entails enabling citizens to prepare online by accessing forms and

**Table 5.** Public service channel strategies summary.

| | Key Findings | Implications | Recommendations |
|---|---|---|---|
| *Digital Channels* | Citizens use online channels for both simple and complex tasks. | Task complexity alone does not determine channel choice; contextual factors matter. | Refine hypotheses by incorporating *task difficulty, ambiguity, and complexity* as variables. |
| *Channel choice* | Offline channels remain widely used despite digital growth. | Citizens prefer in-person interactions for high-uncertainty tasks (Goldsmith & Crawford, 2014). | Avoid forced digital migration; maintain *hybrid service models*. |
| *Channel Strategy* | Online channels are growing but not replacing offline ones. | Governments overestimate digital preference; multi-channel strategies are essential. | Optimize *cross-channel integration* (e.g. 'prepare online, confirm offline'). |
| *Access & Literacy* | Digital divide persists due to access and literacy gaps. | Public access points are less effective than personal device usage (Chen & Dimitrova, 2006). | Invest in *digital literacy programs* and *user-friendly interfaces*. |
| *Channel efficiency* | Channel synergy improves efficiency (Santamaria-Philco et al., 2019). | Seamless interoperability enhances citizen experience. | Implement *unified citizen accounts* and *cross-channel referrals*. |
| *Trust & Culture* | TAM (Venkatesh et al., 2003) explains adoption but needs expansion. | Perceived usefulness and ease of use drive digital adoption, but trust and culture also matter. | Incorporate *trust in government* and *cultural preferences* in future studies. |

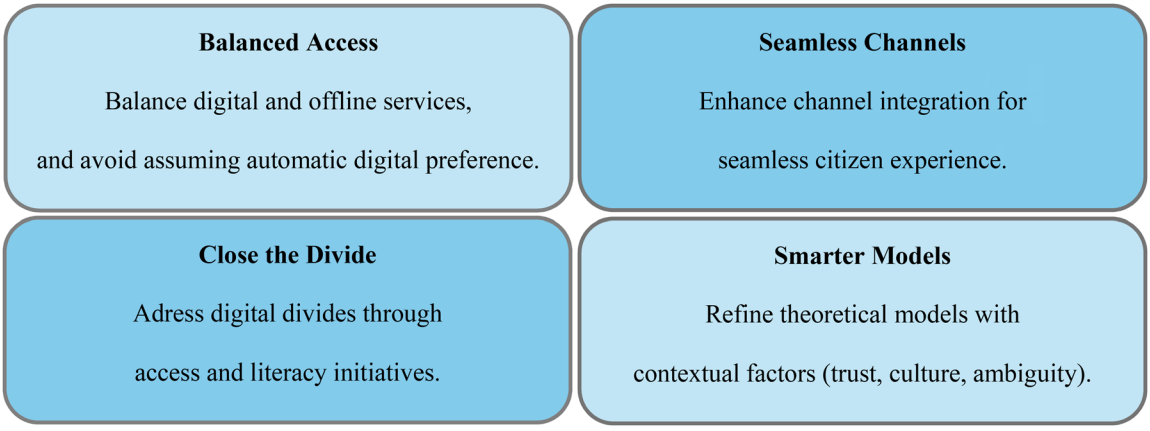



**Figure 7.** Key Takeaways for Policy & Research recommendations.

information digitally before proceeding with face-to-face interactions to seek clarifications and confirmations. Such a model not only streamlines service delivery but also fosters a citizen-centric approach where service design is informed by observable behavioral patterns and preferences.

Exploring the digital engagement landscape reveals a dichotomy between those who possess connected devices and those who do not. The findings suggest that despite the widespread availability of online channels, individuals lacking personal devices continue to utilize these resources via public access points, albeit with a diminished perception of their efficacy. This dichotomy signifies a critical digital divide, an aspect that future policy frameworks must address by focusing not only on device access but also on digital literacy enhancement (Chen & Dimitrova, 2006).

Furthermore, the adoption and adaptation of technology acceptance theories in multichannel management can provide deeper insights into the motivational drivers influencing citizen preferences and behaviors. The Technology Acceptance Model, for instance, offers a theoretical basis for understanding how perceived ease of use and usefulness could sway the adoption of online versus offline channels (Venkatesh et al., 2003). Table 5 sumarize public service channel strategies.

For government agencies in Northwest Tunisia to effectively recalibrate their multichannel strategies, attention must be directed towards an inclusive model that leverages both the efficiency of digital platforms and the reassurance provided by traditional interfaces. Comprehensive multichannel initiatives must go beyond mere service provision, aspiring instead towards fostering an environment where

citizens feel equipped to navigate and choose appropriately from an array of communication conduits. The continuous monitoring and evaluation of these strategies against citizen feedback will ensure alignment with evolving needs, thereby enhancing citizen trust and engagement in public services (Figure 7).

## Conclusion

Summarizing the complex relationships inherent in multichannel management across Northwestern Tunisia's municipalities reveals the importance of an integrated approach to communication channels in improving public service engagements. This study emphasizes the importance of rethinking government service delivery structures, with a focus on integrating both offline and online channels for increased efficiency and inclusivity across varied demographic groups. Such an integrated approach is consistent with broader trends in public administration studies, as observed by Alexopoulos et al. (2019), who argue that combining traditional and digital communication strategies is critical for addressing accessibility gaps.

Santamaria-Philco et al. (2019) explore channel synergy, which advocates for an innovative approach of public service delivery based on cross-channel collaboration. This model addresses and lowers the redundancies found in separate channel topologies, resulting in a seamless service environment that matches with citizens' various preferences and habits (Santamaria-Philco et al., 2019). Furthermore, findings from the Technology Acceptance Model shed light on how perceived relevance and ease of use influence channel selection, lending credence to the idea that well-integrated channels can increase user happiness and engagement.

One important component of this study is the persisting digital gap, which is defined by unequal access to connected devices and necessitates targeted policy initiatives. According to Chen and Dimitrova (2006), overcoming such discrepancies requires not just technological infrastructure upgrades, but also improvements in digital literacy to ensure fair channel access. The fact that persons without personal digital access can still engage online demonstrates adaptive resilience, which needs inclusive digital strategies that accommodate a wide range of access levels and proficiency.

Furthermore, enhancing multichannel management necessitates a cultural shift toward citizen-centric service design. This transition is helped by interactive governance models that encourage a participatory approach and incorporate citizen feedback into service development procedures (Reddick, 2005). By putting citizens at the center of channel strategy design, government bodies can improve service relevance, build trust, and inspire greater involvement.

In conclusion, this study gives critical insights into the complicated environment of multichannel management in public services, laying foundations for future research. By encouraging debate and constant modification, this strategy has the potential to change public sector service delivery into a more responsive, adaptable, and egalitarian system. Such systemic reforms are critical for developing resilient civic institutions that respond to citizens' changing needs and ambitions, assuring long-term involvement and improved public service outcomes in an increasingly digital environment.

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**Akram Jabri**: Conceptualization, Methodology, Software, Data curation, Writing- Original draft preparation, Visualization, Investigation, Writing- Reviewing and Editing. **Professor Mokbul Morshed Ahmad**: Conceptualization, Supervision. All authors have read and approved the final work.

## Authors contributions

CRediT: **Akram Jabri**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Mokbul Morshed Ahmad**: Conceptualization, Supervision.

## Ethical approval

Approval was obtained from the ethics committee of Asian institute of technology AIT Bangkok Thailand. Ref. No.: RERC 2022/019

The procedures used in this study adhere to the tenets of the Declaration of Helsinki.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

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## Data availability statement

Data available on request from the authors: The data that support the findings of this study are available from the corresponding author, upon reasonable request.

## References

Abdelsalam, H. M. E., Reddick, C. G., Gamal, S., & Al-Shaar, A. (2013). Social media in Egyptian government websites: Presence, usage, and effectiveness. *Government Information Quarterly*, *30*(4), 406–416. https://doi.org/10.1016/j.giq.2013.05.020

Alexopoulos, C., Lachana, Z., Androutsopoulou, A., Diamantopoulou, V., Charalabidis, Y., & Loutsaris, M. A. (2019). How machine learning is changing e-government. In S. Ben Dhaou, L. Carter, & M. Gregory (Eds.), *12th International Conference on Theory and Practice of Electronic Governance (ICEGOV 2019)* (pp. 354–363). ACM. https://doi.org/10.1145/3326365.3326412

Bélanger, F., & Carter, L. (2008). Trust and risk in e-government adoption. *The Journal of Strategic Information Systems*, *17*(2), 165–176. https://doi.org/10.1016/j.jsis.2007.12.002

Cai, X. (2008). Jan A. G. M. van Dijk. The deepening divide: Inequality in the information society. Thousand Oaks, CA: Sage, 2005, 240 pp., ISBN 141290403X (paperback). *Mass Communication and Society*, *11*(2), 221–224. https://doi.org/10.1080/15205430701528655

Castilla, R., Pacheco, A., Robles, I., Reyes, A., & Inquilla, R. (2021). Digital channel for interaction with citizens in public sector entities. *World Journal of Engineering*, *18*(4), 547–552. https://doi.org/10.1108/WJE-08-2020-0377/FULL/XML

Chatfield, A. T., & Reddick, C. G. (2018). *Open Data Policy and Open Data Services* [Paper presentation]. 24th Americas Conference on Information Systems (AMCIS 2018) (pp. 1–9). https://aisel.aisnet.org/amcis2018/eGovernment/Presentations/5

Chen, Y.-C., & Dimitrova, D. V. (2006). Electronic government and online engagement. *International Journal of Electronic Government Research*, *2*(1), 54–76. https://doi.org/10.4018/JEGR.2006010104:

Criado, J. I., & Villodre, J. (2018). Public employees in social media communities: Exploring factors for internal collaboration using social network analysis. *First Monday*, *23*(4)[, 1–17. https://doi.org/10.5210/fm.v23i4.8348

Dijk, J. v., & Hacker, K. L. (2018). Internet and democracy in the network society. In *Routledge studies in global information, politics and society*. Routledge.

Field, A. (2013). *Discovering statistics using IBM SPSS statistics*. https://www.akademika.no/okonomi-administrasjon-og-ledelse/statistikk/discovering-statistics-using-ibm-spss-statistics/9781446274583r180

Gamboa Mora, M. C., Vera-Monroy, S. P., Mejía-Camacho, A., & Guerrero Rueda, W. J. (2021). Perception channels and cognitive styles: Opponents, followers or learning allies? *Heliyon*, *7*(2), e06242. https://doi.org/10.1016/J.HELIYON.2021.E06242

Goldsmith, S., & Crawford, S. P. (2014). *The responsive city: Engaging communities through data-smart governance*. https://www.wiley.com/en-au/The+Responsive+City%3A+Engaging+Communities+Through+Data-Smart+Governance-p-9781118910900

Heinhuis, D., & De Vries, E. J. (2009). Modelling customer behaviour in multi-channel service distribution. *Lecture Notes in Business Information Processing*, *23* LNBIP, 47–63. https://doi.org/10.1007/978-3-642-01197-9_4

Helsper, E. J. (2012). A corresponding fields model for the links between social and digital exclusion. *Communication Theory*, *22*(4), 403–426. https://doi.org/10.1111/j.1468-2885.2012.01416.x

Kernaghan, K. (2015). Innovation and public sector service delivery. *Innovation Journal*, *20*(2), 1–18.
Klievink, B., Neuroni, A., Fraefel, M., & Zuiderwijk, A. (2017). Digital strategies in action -A Comparative analysis of national data infrastructure development. In C. C. Hinnant & A. Ojo (Eds.), *18th Annual International Conference on Digital Government Research (dg.o 2017)* (pp. 129–138). ACM. https://doi.org/10.1145/3085228.3085270
Lee, J. (2002). A key to marketing financial services: The right mix of products, services, channels and customers. *Journal of Services Marketing*, *16*(3), 238–258. https://doi.org/10.1108/08876040210427227/FULL/XML
Li, D., Wu, R., & Wang, H. (2019). Management Response to Online Customer Reviews in Distribution Channel: A Perspective of Manufacturers. *Proceedings of the Annual Hawaii International Conference on System Sciences*, *2019-January,* 4743–4752. https://doi.org/10.24251/HICSS.2019.572
Lindgren, I., Madsen, C. Ø., Hofmann, S., & Melin, U. (2019). Close encounters of the digital kind: A research agenda for the digitalization of public services. *Government Information Quarterly*, *36*(3), 427–436. https://doi.org/10.1016/j.giq.2019.03.002
Madsen, C. Ø., & Kræmmergaard, P. (2015). The efficiency of freedom: Single parents' domestication of mandatory e-government channels. *Government Information Quarterly*, *32*(4), 380–388. https://doi.org/10.1016/j.giq.2015.09.008
Madsen, C. Ø., & Kræmmergaard, P. (2016). How to succeed with multichannel management: A case study of cross-organizational collaboration surrounding a mandatory self-service application for Danish single parents. *International Journal of Public Administration in the Digital Age*, *3*(4), 94–110. https://doi.org/10.4018/IJPADA.2016100107
Margetts, H., & Dunleavy, P. (2013). The second wave of digital-era governance: A quasi-paradigm for government on the Web. *Philosophical Transactions. Series A, Mathematical, Physical, and Engineering Sciences*, *371*(1987), 20120382. https://doi.org/10.1098/RSTA.2012.0382
Millard, J. (2007). E-government for an inclusive society: How different citizen groups use E-government services in Europe. *E-Government Research: Policy and Management*, 254–267. https://doi.org/10.4018/978-1-59904-913-7.CH011
Moon, Y. J. (2013). The tangibility and intangibility of e-service quality. *International Journal of Smart Home*, *7*(5), 91–102. https://doi.org/10.14257/ijsh.2013.7.5.10
Noble, S. M., Griffith, D. A., Weinberger, M. G., Noble, S. M., Griffith, D. A., & Weinberger, M. G. (2005). Consumer derived utilitarian value and channel utilization in a multi-channel retail context. *Journal of Business Research*, *58*(12), 1643–1651. https://EconPapers.repec.org/RePEc:eee:jbrese:v:58:y:2005:i:12:p:1643-1651 https://doi.org/10.1016/j.jbusres.2004.10.005
Panagiotopoulos, P., Klievink, B., & Cordella, A. (2019). Public value creation in digital government. *Government Information Quarterly*, *36*(4), 101421–101428. https://doi.org/10.1016/j.giq.2019.101421
Pieterson, W. (2010). Citizens and service channels: Channel choice and channel management implications. In *Citizens and E-Government: Evaluating policy and management* (pp. 250–267). IGI Global Scientific Publishing. https://doi.org/10.4018/978-1-61520-931-6.CH014
Pieterson, W. J., & Ebbers, W. E. (2020). Channel choice evolution: An empirical analysis of shifting channel behavior across demographics and tasks. *Government Information Quarterly*, *37*(3), 101478. https://doi.org/10.1016/j.giq.2020.101478
Pieterson, W., & Ebbers, W. (2008). The use of service channels by citizens in the Netherlands: Implications for multi-channel management. *International Review of Administrative Sciences*, *74*(1), 95–110. https://doi.org/10.1177/0020852307085736
Pieterson, W., Ebbers, W., & Madsen, C. Ø. (2017). New Channels, New Possibilities: A Typology and Classification of Social Robots and Their Role in Multi-channel Public Service Delivery. *16th International Conference on Electronic Government (EGOV 2017),* 47–59. https://doi.org/10.1007/978-3-319-64677-0_5
Pina, V., Torres, L., & Royo, S. (2009). E-government evolution in EU local governments: A comparative perspective. *Online Information Review*, *33*(6), 1137–1168. https://doi.org/10.1108/14684520911011052
Reddick, C. G. (2005). Citizen interaction with e-government: From the streets to servers? *Government Information Quarterly*, *22*(1), 38–57. https://doi.org/10.1016/j.giq.2004.10.003
Reddick, C. G., Abdelsalam, H. M. E., & Elkadi, H. A. (2012). Channel choice and the digital divide in e-government:the case of Egypt. *Information Technology for Development, 18*(3), 226–246. https://doi.org/10.1080/02681102.2011.643206
Riggins, F. J., & Mitra, S. (2007). An E-Valuation Framework for developing net-enabled business metrics through functionality interaction. *Journal of Organizational Computing and Electronic Commerce*, *17*(2), 175–203. https://doi.org/10.1080/10919390701294129
Santamaría-Philco, A., & Wimmer, M. A. (2018). Trust in e-participation: An empirical research on the influencing factors. In In C. C. Hinnant & A. Zuiderwijk (Eds.), *19th Annual International Conference on Digital Government Research: Governance in the Data Age (dg.o 2018)* (pp. 1–10). ACM. https://doi.org/10.1145/3209281.3209286
Santamaria-Philco, A., Canos Cerda, J. H., & Penades Gramaje, M. C. (2019). Advances in e-participation: A perspective of last years. *IEEE Access*, *7*, 155894–155916. https://doi.org/10.1109/ACCESS.2019.2948810
Thomas, J. C., & Streib, G. (2003). The new face of government: Citizen-initiated contacts in the era of E-government. *Journal of Public Administration Research and Theory*, *13*(1), 83–102. https://doi.org/10.1093/jopart/mug010
Tourangeau, R., Conrad, F. G., & Couper, M. P. (2013). The science of web surveys. Oxford University Press. https://doi.org/10.1093/ACPROF:OSO/9780199747047.001.0001

van Deursen, A. J. A. M., & Helsper, E. J. (2015). The third-level digital divide: Who benefits most from being online? In *Communication and information technologies annual (Studies in Media and Communications, Vol. 10)* (pp. 29–52). Emerald Group Publishing Limited. https://doi.org/10.1108/S2050-206020150000010002

van Dijk, J. A. G. M., Pieterson, W., Van Deuren, A., & Ebbers, W. (2007). E-services for citizens: The Dutch usage case. In M. A. Wimmer, J. Scholl, & A. Gronlund (Eds.), *Electronic Government: Proceedings of the 6th IFIP WG 8.5 International Conference, EGOV 2007* (Vol. 4656, pp. 155–166). https://doi.org/10.1007/978-3-540-74444-3

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly: Management Information Systems*, *27*(3), 425–478. https://doi.org/10.2307/30036540

Verhoef, P. C., Doffer, A., & Wilde, E. d (2005). De verdwenen klant: Uitdagingen voor bedrijven in een multi-channel-landschap. *Holland Management Review*, *22*(104), 38–46. https://research.rug.nl/en/publications/de-verdwenen-klant-uitdagingen-voor-bedrijven-in-een-multi-channe

Villodre, J., & Criado, J. I. (2020). Analyzing social media institutionalization in public administration. The role of inhibitors in local governments. In *ACM International Conference Proceeding Series* (pp. 31–40). https://doi.org/10.1145/3396956.3396984

Vives, X. (2019). Digital disruption in banking. *Annual Review of Financial Economics*, *11*(1), 243–272. https://doi.org/10.1146/annurev-financial-100719-120854

Vries, G. D. (2020). Do digital technologies put democracy in jeopardy? In M. Hildebrandt & K. O'Hara (Eds.), *Life and the law in the era of data-driven agency* (pp. 135–155). Edward Elgar Publishing. https://www.elgaronline.com/view/edcoll/9781788971997/9781788971997.00015.xml

West, D. M. (2004). E-government and the transformation of service delivery and citizen attitudes. *Public Administration Review*, *64*(1), 15–27. https://doi.org/10.1111/j.1540-6210.2004.00343.x

Wirtz, B. W. (2022). *The information society and digital government* (pp. 21–42). Springer Texts in Business and Economics. https://doi.org/10.1007/978-3-031-13086-1_2

Wirtz, B. W., Becker, M., & Langer, P. F. (2023). An integrated model of digital open government. *International Journal of Public Administration*, *46*(13), 951–970. https://doi.org/10.1080/01900692.2022.2050386

Wirtz, B. W., & Birkmeyer, S. (2015). Open government: Origin, development, and conceptual perspectives. *International Journal of Public Administration*, *38*(5), 381–396. https://doi.org/10.1080/01900692.2014.942735

Wirtz, B., & Kurtz, O. T. (2016). Citizen preferences toward e-government city portals - An empirical analysis of full online, mobile and social media services. *International Public Management Review*, *17*(1), 1–20.

Yao, Y., & Xu, P. (2022). E-participation decision across different channels. *Information Technology & People*, *35*(3), 956–976. https://doi.org/10.1108/ITP-05-2020-0314/FULL/XML

Yildiz, M. (2007). E-government research: Reviewing the literature, limitations, and ways forward. *Government Information Quarterly*, *24*(3), 646–665. https://doi.org/10.1016/j.giq.2007.01.002